HARDWARE

1.Draw a labeled diagram of a desktop PC, clearly marking the motherboard, CPU, RAM, GPU, and both HDD and SSD storage components.

Ans.

**2. Create a comparison table showing the main differences between HDD and SSD storage in terms of speed, durability, and typical use cases.

Hint: Think about which one is faster, which is more shock-resistant, and where you would usually find each type (laptop, gaming PC, etc.).**

Ans.

Feature	**HDD (Hard Disk Drive)**	**SSD (Solid State Drive)**
Speed	Slow and lagging	Extremely fast
Durability	Breaks easily (fragile)	Strong and shockproof
Typical Use Cases	Large data backups	Laptops and gaming

3. Write a short explanation (2-3 sentences each) describing the role of the CPU, RAM, and GPU in running a game like PUBG or Free Fire on a PC.

Ans. **CPU (The Brain)**

The CPU controls the whole game. It calculates things like where you are moving, when you shoot, and what the enemy is doing.

RAM (The Helper)

RAM holds the game data temporarily so your computer can read it instantly. This stops the game from lagging or freezing while you play.

GPU (The Painter)

The GPU creates all the pictures on your screen. It draws the map, the characters, and the actions smoothly so everything looks good.

4. Imagine you want to upgrade your PC to run the latest version of FIFA smoothly. List which hardware components you would consider upgrading and explain why for each.

Ans. 1. Graphics Card (GPU)

- **Why upgrade:** The GPU is the most important part for a sports game. It draws the grass on the pitch, the players' faces, and the crowded stadiums.
- **The benefit:** A newer graphics card stops the game from stuttering during matches and replay cuts. It makes everything look sharp and run at a very high fluid speed (frame rate).

2. Processor (CPU)

- **Why upgrade:** FIFA has to calculate player intelligence, ball physics, and referee decisions all at the same time.
- **The benefit:** A faster CPU ensures your game doesn't drop frames or slow down when you make quick passes or when many players crowd around the ball.

3. Storage (SSD)

- **Why upgrade:** Modern football games are huge (around 100 GB) and need to load massive amounts of data instantly. If you are still using an old HDD, the game will take forever to load.

  Steam

- **The benefit:** Upgrading to a Solid State Drive (SSD) makes the game boot up in seconds, skips long loading screens, and lets you jump straight onto the pitch.

4. RAM (Memory)

- **Why upgrade:** The game needs a healthy amount of short-term memory to keep things running cleanly in the background. The latest versions of the game recommend at least 12 GB of RAM to perform perfectly.
- **The benefit:** Upgrading your RAM to 16 GB stops the game from crashing or freezing when things get intense on screen.